reimagining engineering education for the ai era

The unified practical learning workspace connecting academic theory, custom AI video lessons, in-browser coding, and 24/7 WhatsApp mentoring into one tactile notebook.

Live Interactive Prototype

WhatsApp Bot Active

STPI Backed

IN DPIIT Registered Startup

NVIDIA Inception Partner



Meet Owlbert — The Official EziLearning AI Mentor

Daily micro-challenges · Pocket doubt-clearing on WhatsApp

the fragmented learning crisis in indian engineering colleges

Over 15+ Lakh engineering graduates emerge yearly across 3,200+ AICTE institutions in India. Yet, colleges face strict NBA Outcome-Based Education (OBE) and NAAC accreditation mandates while 84% of grads lack day-one software engineering readiness.

1. Passive Theory Silos

Traditional university LMS portals deliver static syllabus PDFs and rote exams with zero telemetry on whether students can write clean, modular software.

2. Severe Toolchain Fatigue

Students juggle 6+ disconnected subscriptions: YouTube lectures, LeetCode, GitHub, disparate WhatsApp groups, and external private finishing schools.

3. The 84% Employability Gap

National Employability Reports confirm over 84% of Indian engineering graduates are unemployable in core tech roles without 6–12 months of costly post-college retraining.

84%
Indian Grads Unready for Core Tech Roles

3,200+
AICTE Colleges Facing NBA/NAAC Mandates

15+ Lakh
Annual Indian Engineering Graduates

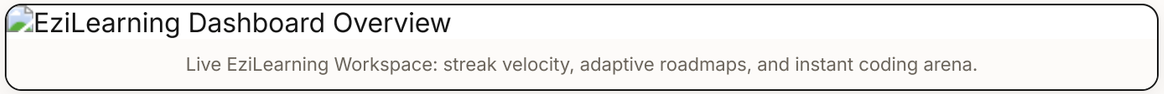
meet ezilearning: one tactile notebook workspace

We unify every stage of the student engineer journey — from first-year fundamentals to recruiter-verified portfolios — into a single, game-inspired tactile interface.

1. Adaptive AI Roadmaps:
Dynamic Directed Acyclic Graph (DAG) that personalizes pacing and prerequisites based on student role targets.

2. AI-Synthesized Video Lessons:
Personalized 90-second Manim/FFmpeg animations generated on-demand to bridge specific code comprehension gaps.

3. 24/7 WhatsApp AI Mentor:
Owlbert the Owl brings 8:30 AM daily 60-second micro-quizzes and instant doubt resolution directly to students' phones.



Live EziLearning Workspace: streak velocity, adaptive roadmaps, and instant coding arena.

customized video lessons tailored to student gaps

Instead of watching static 45-minute generic YouTube tutorials, students receive an AI-generated 90-second visual explanation targeting their exact algorithmic misconception.

1. Telemetry Bottleneck Detection
Identifies off-by-one pointer errors, recursion stack overflow, or memory overhead in coding exercises.

2. Automated Storyboarding & Voiceover
Compiles targeted narration script and programmatic graph scene descriptions in 1.4 seconds.

3. 1080p Local High-Fidelity Rendering
Renders graph nodes, pointer animations, and step-by-step proofs synchronized with voice explanations.

✓ Proven 3.2x higher concept retention vs. passive video lectures



Varin Video Player: AI-generated Dijkstra algorithm relaxation animation (Frame captured at 1:58).

pocket mentoring on whatsapp with owlbert 🦉

Students spend hours away from computers. EziLearning builds daily learning habits by bringing quizzes and doubt-clearing directly to WhatsApp — with zero friction or app install required.

🌞 Daily 8:30 AM 60-Second Micro-Challenges

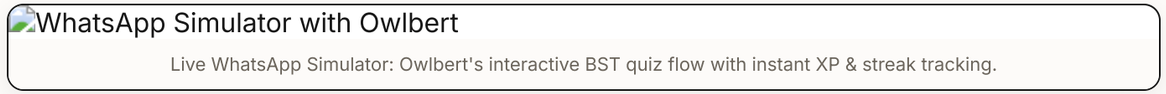
Owlbert sends a bite-sized algorithmic question to students' WhatsApp every morning. Quick taps (A, B, C, D) award XP, keep streaks ablaze, and compound daily retention.

💬 Instant Photo & Voice Doubt Clearing

Stuck during a lecture or late at night? Students snap a picture of code or record a voice query. Owlbert explains edge cases and complexity Socratically right inside chat.

🔗 QR-Code Pairing to Institutional Profile

One-click WhatsApp linking securely synchronizes quiz answers, streak milestones, and telemetry back to the student's campus LMS passport.

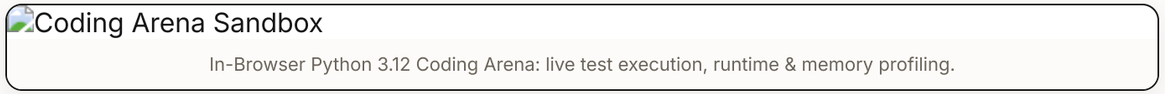


WhatsApp Simulator with Owlbert

Live WhatsApp Simulator: Owlbert's interactive BST quiz flow with instant XP & streak tracking.

zero-friction coding & recruiter-verified proof

We move students from memorizing algorithm solutions to designing production-ready systems with instantaneous runtime benchmarks and verified live portfolio pages.



In-Browser Python 3.12 Coding Arena: live test execution, runtime & memory profiling.

Zero-Install Cloud Sandbox:
Executes Python, C++, Java, and TypeScript directly in the browser with automated test runners and memory profiling.

Architectural Project Milestones:
Students build full-stack systems with step-by-step code checkpoints, pull request evaluations, and schema migrations.

Verified Recruiter Portfolio URLs:
Generates cryptographic proof of code authorship, solving time, and platform contest ratings on one public link.

campus sprint leagues & holistic readiness scoring

Replacing arbitrary 0–100% course completion bars with an industry-calibrated Readiness Index and weekly inter-college coding leagues.

The 5-Factor Readiness Index (0–100%):

1. Algorithmic Mastery	30% Weight
2. Project Architecture & Commits	25% Weight
3. WhatsApp Streak Consistency	20% Weight
4. AI Mock Interview Technical Communication	15% Weight
5. Live Job Market Skill Alignment	10% Weight



Roadmap DAG: dynamic skill nodes, prerequisite unblocking, and verified milestone badging.

a ₹28,000+ cr ($3.4b) indian engineering workforce opportunity

Engineering institutions across India face rigorous NBA Outcome-Based Education (OBE), NAAC accreditation, and NIRF ranking mandates while tech employers struggle with candidate interview noise.

₹28,500 Cr

Total Addressable Market

Indian engineering skilling, college LMS modernization, and tech talent recruitment pipelines (22% CAGR).

3,200+

AICTE Engineering Colleges

Accredited institutions seeking automated coding labs, OBE telemetry, and verified placement records.

15+ Lakh

Annual Engineering Grads

Undergraduates requiring daily algorithmic practice, project guidance, and placement support.

Why Now? AI Enablement & Frontier Tech Infrastructure

Backed by **STPI** and partnered with the **NVIDIA Inception program**, Haptix Solution leverages accelerated computing and generative pipelines (Manim/FFmpeg + conversational AI) to deliver personalized video tutoring and 24/7 WhatsApp mentoring at **near-zero marginal cost** across Indian campuses.

backed by premier institutions to transform indian tech education

Haptix Solution Private Limited focuses on making advancements in AI enablement and frontier tech, partnering with elite tech ecosystems to solve India's engineering employability divide.

STPI Backed

Incubated and supported by **Software Technology Parks of India (STPI)**, Ministry of Electronics & IT.

IN

DPIIT Registered

Recognized Indian startup by the **Department for Promotion of Industry and Internal Trade (DPIIT)**.

NVIDIA Inception

Selected partner in the **NVIDIA Inception Program** for frontier AI acceleration and GPU enablement.

Frontier Tech & AI

Proprietary video generation (Manim/FFmpeg), real-time sandboxes, and WhatsApp conversational AI agents.

partner with haptix solution private limited

Roll out EziLearning across your campus. Transform student consistency, NBA/NAAC telemetry, and placement outcomes.

Live Demo:
ezilearning.haptix.in